Connecting to a Private Subnet Using a Jump Server or Bastion Host

To connect to resources in a private subnet that lacks direct internet connectivity, we typically use a **Jump Server** (also known as a **Bastion Host**). This server is placed in a public subnet, allowing secure access to the private subnet via SSH or RDP.

Internet Connectivity for Private Subnet

Since the private subnet does not have direct internet access, we can use a **NAT Gateway** to enable outbound internet connectivity. The NAT Gateway, placed in the public subnet, acts as an intermediary that allows instances in the private subnet to initiate outbound connections to the internet while keeping their private IP addresses hidden from external sources.

What is a NAT Gateway?

A **NAT Gateway** (Network Address Translation Gateway) is a service that enables instances in a private subnet to connect to the internet or other AWS services, but it prevents the internet from initiating connections with those instances. The NAT Gateway translates the private IP addresses of the instances to a public IP address for outbound traffic, masking the identity of the private subnet.

Key Points to Remember:

- **NAT Gateway** is deployed in a public subnet.
- It allows instances in private subnets to access the internet while keeping them isolated from direct inbound internet traffic.
- The NAT Gateway uses an Elastic IP address for outbound connections, ensuring the private subnet remains secure.
- **Jump Server/Bastion Host** is used to securely access resources in a private subnet.